

MOTIONCORE · DESIGN DECISIONS REGISTER

MotionCore design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/motioncore		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open items from REVIEW.md, MTC-DDR-001 to 003 and the build work; budget treated as a value-engineering target

MotionCore design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Design for construction: accept the changes that make the concept buildable (finned tube on a floor plate, rivet nuts, sockets through the wall, supervisor board layout, bench rig and the others)	Accept; change some; revert	Accept	The whole build plan	MTC-DDR-003, Table 1
2	Interface v0.2: a fifth socket for the speed sensor and the pin-out of all sockets	(a) M8 4-pin speed socket and the pin-out of build plan Table 3; (b) a 12-pin M12 safety loop socket that also carries the speed sensor	(a)	Tube drilling, sockets, harness, supervisor wiring	MTC-DDR-003, A1
3	Bench rig for the first prototype	(a) build it, outside the kit total (USD 35); (b) mount the first prototype straight on CargoMule	(a)	Bench board, uprights, feet, posts, handlebar	MTC-DDR-003, A2
4	Contacting coil economizer (holds the coil at about 1 W)	Fit; do not fit	Fit: the heavy case falls from 58.4 to about 55.6 °C on the walls and from 60.0 to about 57 °C on the floor plate, which the floor joint brought to the R9 limit	Supervisor coil switch and wiring	MTC-DDR-002 item 19; MTC-DDR-003, A3
5	Which sealed power connector replaces the unsealed XT90 (R10)	Candidates from the evaluation decided in MTC-DDR-002 item 18	None yet; the evaluation is a paper task still to do	Power socket hole and frame; the prototype is built with the XT90	MTC-DDR-002 item 18
6	Dual-motor hosts (PalletPilot)	Two controllers on one supervisor; two modules	None made	Not part of the first prototype	MTC-DDR-001 item 12
7	Brushed motors (StepClimber)	VESC DC mode; a different power stage	None made	Not part of the first prototype	MTC-DDR-001 item 13
8	Appearance model differences from <code>model.py</code> (lid window, ready and fault lights, two-flange hub, render layout, simplified harness)	Accept each as render-only, or add to the design	Accept as render-only; renders to be redone on the constructable design	None (renders only)	<code>docs/REVIEW.md</code> , session of 2026-09-26

#	Decision needed	Options	Recommendation	Affects in the build	Source
9	Formal review of the license boundary between the VESC firmware (GPL-3.0) and the MIT supervisor firmware	Review now; review before release	Before release	None	MTC-REQ-001 R13

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	A finned enclosure extrusion near 220 x 140 mm with 3 mm walls, vertical-running fins about 3 x 14 mm on a 19 mm pitch and four inside-corner screw ports; if none is stocked, a short custom run or a machined body	Sets the tube, the floor plate's corner holes and the thermal figures	MTC-DDR-003, P1
2	The controller's mounting-hole pattern, flat base for the thermal pad, bus capacitance (about 1,000 μ F) and MOSFET resistance	Sets the floor plate's controller holes and the precharge and thermal figures	MTC-CAL-001 section 1
3	The contactor's foot holes, release time (50 ms or less with the Zener) and DC breaking rating (50 A at 60 V)	Sets the floor holes and the e-stop timing	MTC-CAL-001 section 5
4	The reference motor's axle width (about 135 mm), axle flats (10 mm) and six-bolt disc mount	Sets the upright spacing and slot, and the magnet ring	MTC-DDR-003, P10, P11
5	Brake levers with normally closed (not normally open) cut-off switches	The interlock expects a closed contact when released	MTC-PRC-001 item 10
6	The M12 sockets come with 20 mm four-screw square flanges and the XT90 with a two-screw panel frame	Sets the end wall holes	MTC-DDR-003, P5
7	The prospective short-circuit current of a CellGuard 16S pack against the fuse's 1 kA breaking capacity	Fuse choice for CellGuard hosts	MTC-CAL-001 section 3
8	Whether a SwapCell pack in legacy discharge accepts a heartbeat and moves to mode 2 without opening its output	The supervisor is powered from the pack	docs/REVIEW.md , cross-repo actions

■ Value engineering

Value-engineering target: USD 300 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 293 for the MotionCore kit, items 2 to 14 (USD 7 under the target). The reference hub motor (USD 70) is costed to each host, and the bench rig (USD 35) belongs to the prototype only. Main cost drivers and savings worth trying:

- The largest lines are the controller (USD 85), the enclosure tube and floor plate (USD 30), the supervisor (USD 30), the contactor (USD 25) and the connector set (USD 24).
- Making the design constructable added USD 28: the enclosure tube and floor plate (USD 8 more), the sockets with flanges and frames, the M8 speed socket and the membrane vent (USD 6), brake levers in place of bare

switches (USD 6), the speed sensor's plug (USD 2), the harness branch (USD 2) and rivet nuts and fixings (USD 4).

- Savings worth trying: a stock finned extrusion close to the drawn size rather than a custom one; buying the motor, the brake levers and the pod as one e-bike kit; and, once the interface is settled, laying the supervisor out on one small circuit board in place of modules on a carrier board (TRL 4 work).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items 1 to 11: budget scope (motor costed to hosts), 20 V lower bound, VESC-class controller, DC contactor, stop category 0, interface v0.1, SwapCell heartbeat, CellGuard link, 350 W heavy case, separate firmware, CargoMule as first host	Amish: "i accept all your recommendations, go with them across all repos."	MTC-DDR-001, MTC-DDR-002
2026-09-25	R12 module mass limit relaxed from 1.5 to 2.0 kg	Amish, same instruction	MTC-DDR-002 item 15
2026-09-25	TRL 3 engineering rules: two CAN buses, 0.6 s precharge with a 90 % bus check, DC-rated fuses with 1 kA breaking capacity, 4 mm ² leads for 24 V hosts, 24 V Zener coil suppressor	Amish, same instruction	MTC-DDR-002 item 16
2026-09-25	R1 upper bound raised from 58 to 60 V for CellGuard 16S packs (overvoltage fault 66 V, fuses 60 V DC as consequences)	Amish, same instruction	MTC-DDR-002 item 17
2026-09-25	Evaluate a sealed power connector before interface v0.1 is frozen; XT90 provisional	Amish, same instruction	MTC-DDR-002 item 18
2026-10-01	The budget is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	<code>.kit/STANDARDS.md</code> section 18